

Campus Food Web

Prototype turn guide and complete card text

Game at a Glance

Players: 2-4 **Deck:** 40 organism cards + 8 effect cards **Goal:** First player to reach 15 points.

Each player controls a personal **Garden Plot**, but every plot is part of one connected ecosystem. Organisms may pollinate, feed, hunt, or otherwise interact across the whole table.

Scoring

Check your score at the end of your turn. Scores are not cumulative; your garden must currently be worth 15 points.

- **1 point** per living organism card in your plot.
- **+1 point** for each organism in your plot with 2 or more Population.
- **+1 Biodiversity Bonus** if your plot contains at least one Plant, Herbivore, Predator, and Decomposer.

One Counter Type

All counters are physically identical. Placement determines what they mean:

Counter location	Meaning
On an organism card	Population
Above a plant card	Light
Below a plant card	Water
Beside a plant card	Nutrients
Shared Compost area	Dead organic matter

Every organism enters with **1 Population**. Maximum Population is **3** unless a card says otherwise. Whenever Population is lost, move that counter to Compost. If an organism loses its final Population, discard its card.

Setup

1. Separate the organism cards and effect cards, then shuffle them together into one deck.
2. Give each player one random Plant in their Garden Plot with 1 Population.
3. Each player draws until they have 4 cards in hand.
4. Create a shared counter supply, Compost area, and discard pile.
5. Choose a starting player.

Round Guide

1. Weather

Roll one die once for the whole garden.

1-2 SUNNY: Put 2 Light above every plant. Remove 1 stored Water from every plant because of evaporation.

3-4 PARTLY CLOUDY: Put 1 Light above every plant. No evaporation.

5-6 CLOUDY: Put no Light above plants. No evaporation.

2. Water

Each player takes **3 counters** from the supply and distributes them as Water below plants in their own Garden Plot. Water may remain from previous rounds.

3. Plant Growth

Each plant may spend **1 Light + 1 Water** to gain **+1 Population**.

A plant that also spends **1 Nutrient** may gain **+1 additional Population**. Population still cannot exceed the card's maximum.

4. Player Turns

Beginning with the starting player, each player takes **one turn**.

Draw 1 card.

Play up to 1 card. This may be an organism or an Effect card.

Activate 1 organism you control. A newly played organism may activate immediately.

Turn an activated organism sideways. Each organism can activate only once per round.

5. Cleanup

Return all unused Light to the supply. Straighten activated cards. Water and Nutrients remain in place. Pass the starting-player marker clockwise and begin the next round.

Ecosystem Actions

Pollinate - Mutualism

Choose a compatible flowering plant anywhere in the garden.

Pollinator +1 Population

Plant +1 Population

Both species benefit. Neither may exceed its maximum Population.

Feed - Herbivory

Choose a compatible plant anywhere in the garden.

Herbivore +1 Population

Plant -1 Population

Move the lost plant Population counter to Compost.

Hunt / Parasitize - Predation

Choose compatible prey anywhere in the garden.

Predator +1 Population

Prey -1 Population

Move the lost prey Population counter to Compost.

Decompose

Move **1 counter from Compost** to the Nutrient position beside one of your plants. The Decomposer gains **+1 Population**.

Playing Organisms

Plants can always be played. An animal can be played only if at least one organism it can use for food, prey, or pollination already exists somewhere in the shared garden. Decomposers can always be played.

Playing Effect Cards

An Effect card is played during the “Play up to 1 card” step **instead of an organism**. Resolve it immediately, then place it in the discard pile. Effect cards never score points.

Organism Cards - Plants

Aquatic Milkweed	4 cards
<i>Asclepias perennis</i>	
PLANT - FLOWERING	
Pollinators: Monarch Butterfly, Eastern Carpenter Bee Herbivores: Monarch Butterfly, Aphids	
Host Plant: Monarchs may use Caterpillar Feeding only on Milkweed.	
Frogfruit	3 cards
<i>Phyla nodiflora</i>	
PLANT - FLOWERING	
Pollinators: Monarch Butterfly, Eastern Carpenter Bee	
Drought Tolerant: Sunny weather does not remove stored Water from Frogfruit.	
Blue Mistflower	3 cards
<i>Conoclinium coelestinum</i>	
PLANT - FLOWERING	
Pollinators: Monarch Butterfly, Eastern Carpenter Bee Herbivores: Aphids	
Texas Bluebonnet	3 cards
<i>Lupinus texensis</i>	
PLANT - FLOWERING	
Pollinator: Eastern Carpenter Bee Herbivore: Aphids	
Common Sunflower	3 cards
<i>Helianthus annuus</i>	
PLANT - FLOWERING	
Pollinators: Monarch Butterfly, Eastern Carpenter Bee Herbivore: Aphids	
Large Plant: Sunflower may hold up to 4 Population .	

Organism Cards - Pollinators & Herbivores

Monarch Butterfly	4 cards
<i>Danaus plexippus</i> POLLINATOR - HERBIVORE Choose one action: Pollinate: Visit Milkweed, Frogfruit, Blue Mistflower, or Sunflower. Monarch +1 Population; plant +1 Population. Caterpillar Feeding: Feed on Milkweed. Monarch +1 Population; Milkweed -1 Population.	
Eastern Carpenter Bee	3 cards
<i>Xylocopa virginica</i> POLLINATOR Generalist Pollinator: May Pollinate any flowering plant in the garden.	
Aphids	3 cards
<i>Aphis spp.</i> HERBIVORE Feed: Milkweed, Blue Mistflower, Bluebonnet, or Sunflower. Aphids +1 Population; plant -1 Population. Honeydew Mutualism: If Aphids and Red Imported Fire Ants are in the same Garden Plot, whenever these Aphids successfully Feed, the Fire Ants also gain +1 Population. While protected by Fire Ants, these Aphids cannot be hunted by Milkweed Assassin Bugs or Aphid Parasitoid Wasps. Brown Anoles and Carolina Wrens can still hunt them.	

Organism Cards - Predators

Milkweed Assassin Bug	2 cards
<i>Zelus longipes</i> PREDATOR Hunt: Aphids, Monarch Butterflies, Eastern Carpenter Bees, or Aphid Parasitoid Wasps anywhere in the garden. Cannot hunt Aphids protected by Red Imported Fire Ants.	
Aphid Parasitoid Wasp	2 cards
<i>Lysiphlebus testaceipes</i> PREDATOR - PARASITOID Parasitize: Hunt Aphids anywhere in the garden. Specialist: Cannot hunt other organisms. Cannot hunt Aphids protected by Red Imported Fire Ants.	

Red Imported Fire Ant

2 cards

Solenopsis invicta

PREDATOR - MUTUALIST

Honeydew Mutualism: Protect Aphids in your Garden Plot. Whenever those Aphids successfully Feed, Fire Ants gain +1 Population.

Hunt: Monarch Butterfly, Eastern Carpenter Bee, Milkweed Assassin Bug, or Aphid Parasitoid Wasp.

Brown Anole

2 cards

Anolis sagrei

PREDATOR

Hunt: Any insect anywhere in the garden.

Fire Ant protection does not protect Aphids from Brown Anoles.

Carolina Wren

2 cards

Thryothorus ludovicianus

PREDATOR

Hunt: Any insect or Brown Anole anywhere in the garden.

Organism Cards - Decomposers

Turkey Tail Fungus	2 cards
<i>Trametes versicolor</i> DECOMPOSER - FUNGUS Decompose: Move 1 Compost counter to the Nutrient position beside one of your plants. Turkey Tail gains +1 Population.	
Soil Bacterium	2 cards
<i>Bacillus subtilis</i> DECOMPOSER - BACTERIUM Decompose: Move 1 Compost counter to the Nutrient position beside one of your plants. Rapid Recycling: At 2+ Population, it may instead move 2 Compost to 2 Nutrients with one activation.	

Effect Cards

Rainstorm	2 cards
<i>Effect card</i> EFFECT - play instead of an organism Add 1 Water below every plant in the garden.	
Drought	1 card
<i>Effect card</i> EFFECT - play instead of an organism Remove 1 Water from every plant in the garden that has Water.	
Heat Wave	1 card
<i>Effect card</i> EFFECT - play instead of an organism Remove 1 Water from every plant in the garden. Then choose one plant you control; it gains +1 Population if it still has at least 1 Water remaining.	
Compost Addition	1 card
<i>Effect card</i> EFFECT - play instead of an organism Add 2 counters from the supply directly to the shared Compost area.	

Mulch	1 card
<i>Effect card</i> EFFECT - play instead of an organism Choose one plant you control. Place this card under it until the start of your next turn. Sunny-weather evaporation does not remove Water from that plant. Then discard Mulch.	
Native Plant Boom	1 card
<i>Effect card</i> EFFECT - play instead of an organism Choose one native plant you control. It gains +1 Population , up to its maximum.	
Pesticide Spray	1 card
<i>Effect card</i> EFFECT - play instead of an organism Choose one Garden Plot. Every insect in that plot loses 1 Population . Move all lost Population to Compost.	

Prototype Interaction Reference

Organism	Can interact with
Milkweed	Monarch, Carpenter Bee, Aphids
Frogfruit	Monarch, Carpenter Bee
Blue Mistflower	Monarch, Carpenter Bee, Aphids
Texas Bluebonnet	Carpenter Bee, Aphids
Sunflower	Monarch, Carpenter Bee, Aphids
Monarch	Milkweed; several flowering plants; predators
Carpenter Bee	Any flowering plant; predators
Aphids	Four plants; Fire Ants; multiple predators
Assassin Bug	Aphids, Monarch, Carpenter Bee, Parasitoid Wasp
Parasitoid Wasp	Aphids
Fire Ant	Aphids; several insect prey
Brown Anole	Any insect
Carolina Wren	Any insect; Brown Anole
Turkey Tail	Compost -> Nutrients
Bacillus subtilis	Compost -> Nutrients

Prototype Notes

- The main strategic goal is to build a high-scoring plot while using the entire shared garden for ecological interactions.
- Pollination is intentionally powerful because it demonstrates mutualism: both populations increase.
- The Aphid-Fire Ant relationship demonstrates a different mutualism: aphids provide honeydew while ants protect them from some natural enemies.
- Population loss always creates Compost, so predation and herbivory feed the decomposer-nutrient-plant cycle.
- For the first playtest, record which cards are hard to play, which organisms reach 3 Population too easily, and how many rounds it takes to reach 15 points.